

Sharknado Probability Risk Analysis

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1 Preface

You are a data scientist for a mid-sized business, in a small group of 3-4 data scientists. You've been tasked with creating a report evaluating a scenario for your business. Your colleagues will also be evaluating the same scenario, and your reports will be used in aggregate to determine a consensus (or lack thereof) on the company's action. The reports will also be used to inform downsizing that is rumored to be coming - you want to ensure your report is better than your peers so that you aren't as easy to cut.

You may talk to your peers who are assigned the same scenario, but you do not want to collaborate too closely, lest you both become targets of the rumored layoffs.

I've scaffolded this report for you to make this process easier - as we talk about different sections of a report in class and read about how to create similar sections, you will practice by writing the equivalent section of your report.

The basic steps for this task are as follows:

- Identify the research question from the business question
- Identify data set(s) which are (1) publicly available (you don't have a budget to pay for private data) and (2) relevant to your task
 - (HW Week 6) Document your data sets in `draft-data-doc.qmd`
- Conduct a statistical analysis to support your answer to your research and business questions
 - Write a methods section for your business report corresponding to your statistical analysis
 - (HW Week 9) Draft of results section of business report with relevant graphics/visual aids in `draft-results.qmd`
- Write your report
 - (HW Week 10) Draft of Intro/Conclusion sections in `draft-intro-conclusions.qmd`

- (HW Week 11) Draft of Executive summary section in `draft-exec-summary.qmd`
- Revise your report
 - (HW Week 12 – not turned in) Revise your report
 - (HW Week 13) - Rough draft of report due. Create one or more qmd files for your report (you can overwrite or delete `intro.qmd` and `summary.qmd`), include the names of each file (in order) in `_quarto.yml`. You should use references (edit `references.bib` and use pandoc citations). Make sure your report compiles and looks reasonable in both html and pdf.
 - Develop a presentation to go along with your report (Week 13). Create slides for your report using quarto.
- Peer revise reports
 - Peer revise reports
 - (HW Week 14) - Make edits to your report from comments received from peer review
- Final report & presentation due

2 Executive Summary

This report estimates the probability of a “sharknado” by incorporating real-world constraints and combining the probabilities of the events required for such an occurrence. Using available tornado data and reasonable assumptions, the analysis evaluates the likelihood of three key components:

- 1) The presence of a shark on the coast
- 2) A tornado moving coastal to inland
- 3) A tornado is strong enough to lift and move a shark inland

The methodology excludes tornadoes with missing ending coordinates, as these observations show no bias with respect to geographic location or tornado magnitude. The analysis then sequences the relevant probabilities to estimate the overall likelihood. A key assumption is that sharks are always present when a coastal-to-inland tornado occurs, which intentionally overestimates the probability and provides a conservative upper bound.

Overall likelihood of what? May tilt the focus of the paper being “given a shark is there, do tornadoes have the power to lift them?” The assumption of a shark being always present seems to invalidate the first key component

The results indicate that, even under favorable assumptions, the probability of a sharknado is effectively zero. This supports the conclusion that sharknados are purely fictional and pose no real-world risk.

! Important

The probability of a sharknado is effectively zero. As such, no adjustments to insurance products are warranted. Resources should instead focus on higher-probability catastrophic events such as hurricanes and standard tornadoes.

3 Introduction

Extreme weather events have long piqued public and scientific interest, especially when their consequences can be extremely dangerous for human life. Among these, the concept of a “sharknado,” a tornado capable of lifting sharks from the ocean and throwing them inland, represents an exaggerated but intriguing scenario. While such an event is widely seen as fictional, evaluating a sharknado’s plausibility provides an opportunity to apply probabilistic reasoning within real-world constraints.

Maybe make a reference to how it was popularized with the movie? How has it piqued public and scientific interest (references opportunity here)?

This project aims to estimate the probability of a sharknado occurring by breaking up the phenomenon into a sequence of events that need to occur. In the methods section, we outline the sequence of these probabilities. These include the probability of a tornado forming near a coastal region and moving inland, the probability that such a tornado reaches a sufficient intensity to pick up and move a shark, and the assumption that a shark is present at the exact location required. To ensure a conservative upper estimate for this probability, several assumptions are intentionally biased in favor of the event occurring, such as a constant availability of sharks along coastlines and the ability for high magnitude earthquakes to pick them up and transport them inland.

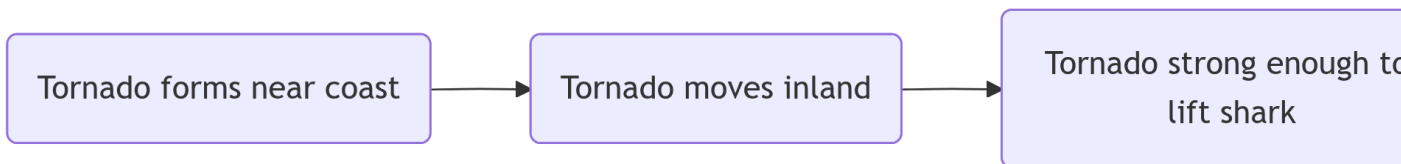


Figure 3.1: Flowchart illustrating the sequence of events leading to a Sharknado.

Maybe mention more explicitly why you are focused on the tornado moving inland since a tornado could still be a sharknado even if it doesn’t go inland. Also what about tornadoes that do both? A tornado could go off the coast and then back on as their paths are never perfectly linear

Using a combination of probabilistic reasoning and simplifying assumptions, the analysis synthesizes these individual components into an overall estimate for the probability of the occurrence of a sharknado. These results provide insight not only into the probability of a sharknado, but also into how compounding rare events together quickly diminishes the overall probability.

4 Methods

maybe introduce this section better and the table? Seems outta place, like a quote bomb but a table bomb

A table summarizing the tornado data from the NOAA and their possible values is below:

In the original US tornado data set from the NOAA, many tornadoes have missing ending latitudes and longitudes, specifically they were marked as “0” instead of “NA”. This can’t be right however since the 0 latitude line and the 0 longitude line do not intersect the US anywhere. The red dotted lines below show 0 latitude and 0 longitude.

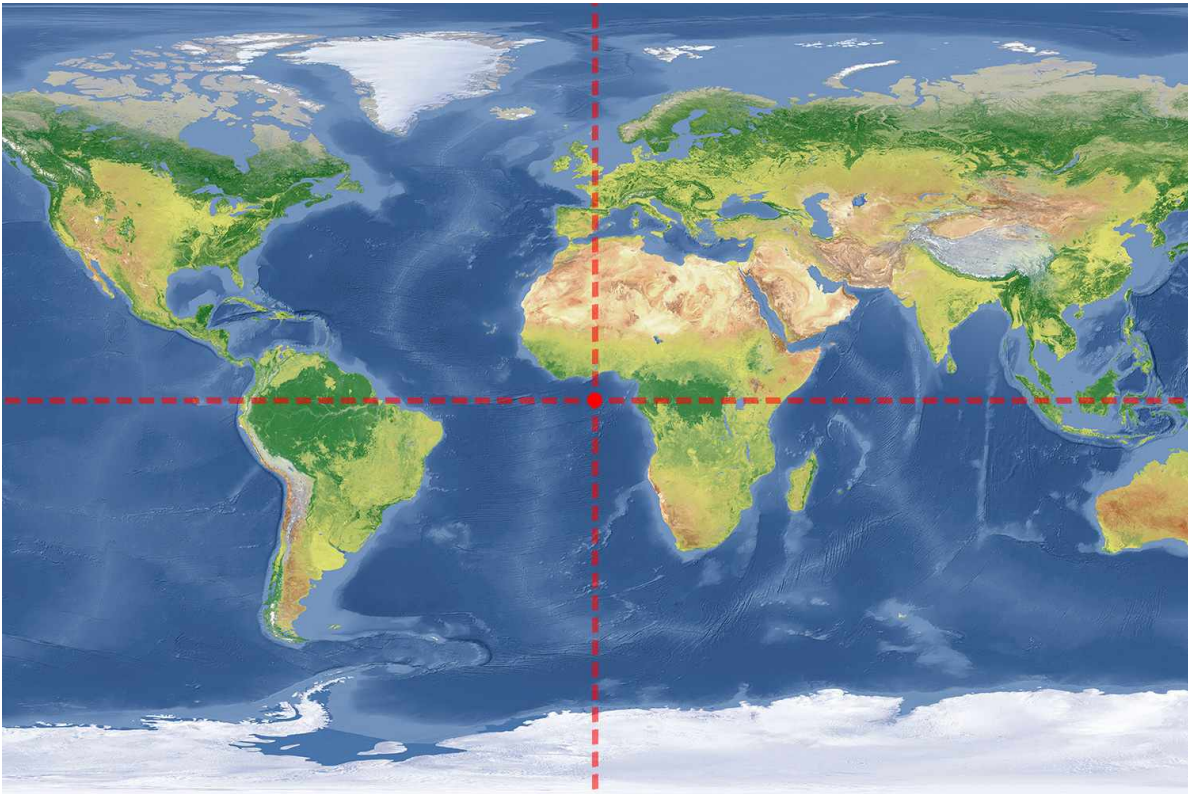


Figure 4.1: Figure: red lines showing 0 latitude and 0 longitude on a map

Tornado Data Variable Descriptions

Variable Name	Description	Type	Possible Values
yr	Year Tornado Occurred	Integer	1950-2024
date	Exact Date Tornado Occurred	Date	01-01-1950 to 12-31-2024
st	State Abbreviation Where Tornado First Touched Down	Categorical	Any US State
mag	EF Scale Magnitude of the Tornado	Integer	0-5 or -9 M
slat	Latitude Where the Tornado First Touched Down	Numeric	US Latitude
slon	Longitude Where the Tornado First Touched Down	Numeric	US Longitude
elat	Latitude Where the Tornado Ended	Numeric	US Latitude
elon	Longitude Where the Tornado Ended	Numeric	US Longitude
len	Tornado Path Length (Miles)	Numeric	0-250 Miles
wid	Tornado Path Width (Yards)	Numeric	0-3500 Yards

As such, these tornadoes in the data set are taken to be rows with missing values. It should be examined whether these tornadoes have some sort of trend relative to the other tornadoes in the data set, so that it can be determined if they can be removed without worry. Specifically, there are two possible concerns: whether tornadoes in a certain area have a significantly higher chance of missing end values, and whether tornadoes of different magnitudes are more likely to have missing end coordinates than others. To examine the first concern, two figures can be created. One of these shows the location of all tornadoes in the data set, and the other shows the location of only the tornadoes that have a missing value for the ending latitude or longitude.

There is no apparent trend in the missing end coordinates. Their geographic distribution is similar to that of all tornadoes, suggesting that the first concern is alleviated.

To address the second concern regarding magnitude, we can create plots for each magnitude level to examine whether the rate of missing end coordinates varies by magnitude, when compared to the overall pattern in the full dataset.

Note

There are no tornadoes with a recorded magnitude of -9 (unknown) that also have missing end coordinates.

From these plots, it appears that higher magnitude tornadoes occur more frequently in the Midwest and South and less often on the West Coast. However, this pattern is consistent with the plots of all tornadoes separated by magnitude. Therefore, while there is some variation by magnitude, excluding tornadoes with missing end coordinates should not heavily bias the analysis, as the trend in magnitude are similar between the full dataset and the subset with missing end coordinates.

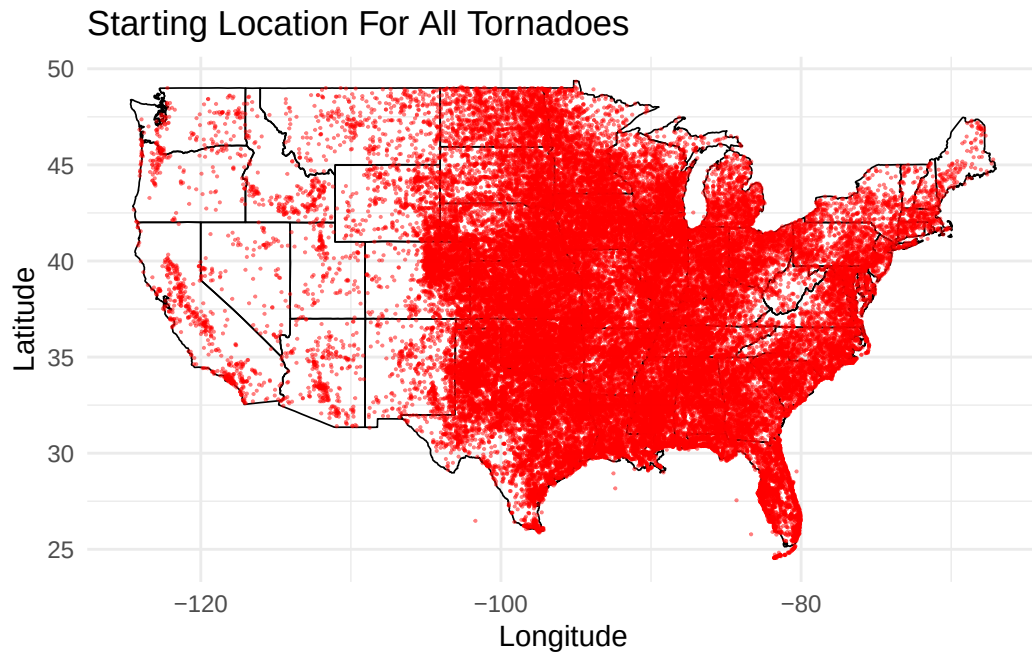


Figure 4.2: Starting locations of all tornadoes in the continental United States.

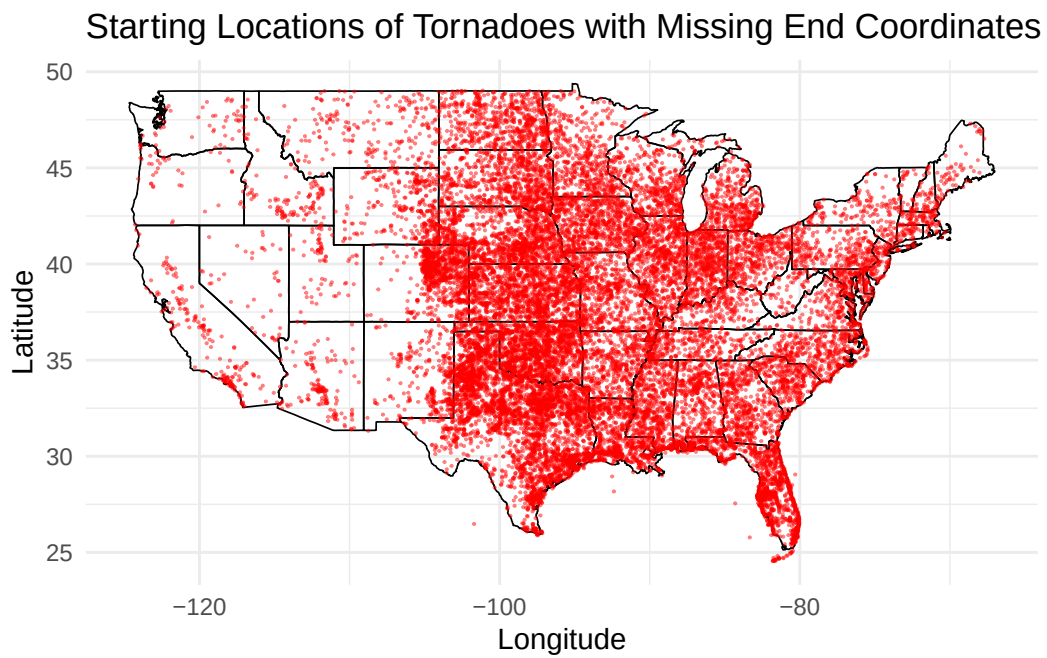


Figure 4.3: Starting locations of tornadoes with missing end coordinates.

Starting Location All Tornadoes Grouped by Magnitude

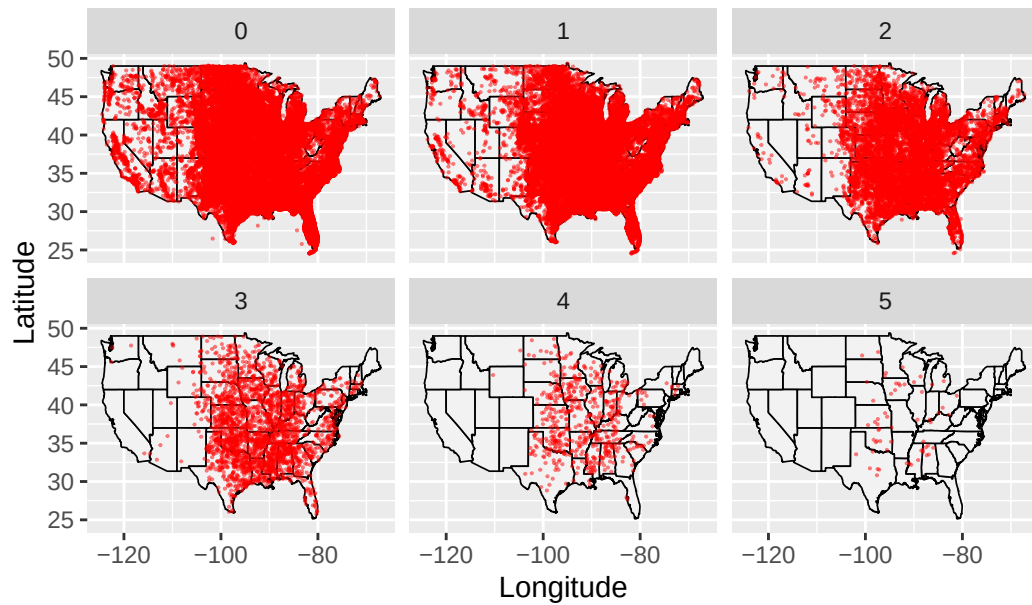


Figure 4.4: Starting locations of all tornadoes in the continental United States grouped by magnitude on EF scale

Starting Location for Tornadoes with Missing End Coordinates (

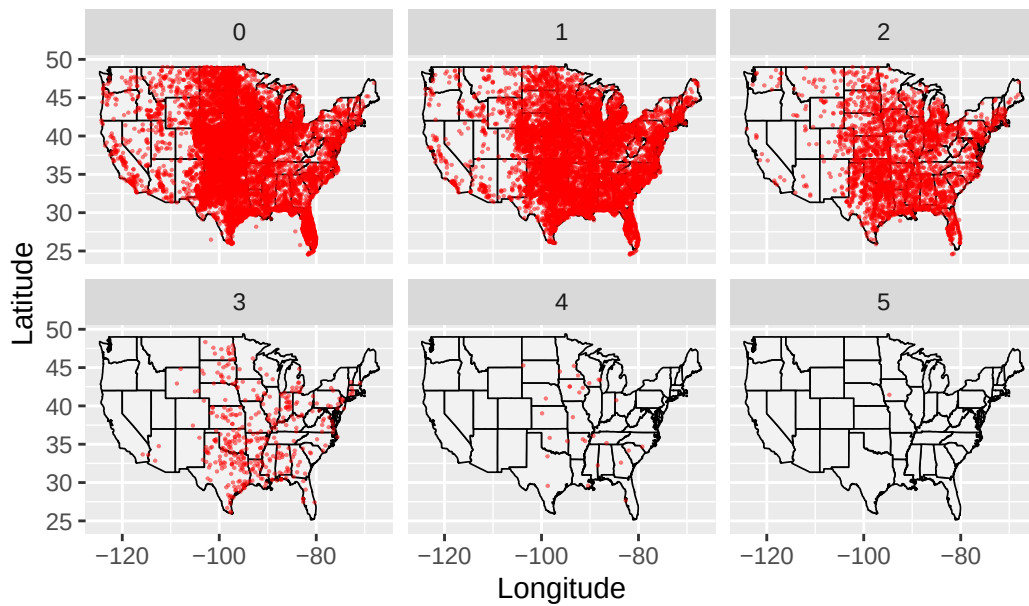


Figure 4.5: Starting locations of tornadoes with missing end coordinates grouped by magnitude on EF scale

To examine the potential risk of a sharknado, we compute the probability of such an event occurring. From the perspective of any tornado occurring in the United States, this probability can be expressed as the product of several individual probabilities:

$$P(\text{sharknado}) = P(\text{shark is present}) \times P(\text{tornado is coastal to inland}) \times P(\text{tornado is strong enough to lift a shark})$$

Alternatively, if we focus specifically on tornadoes that are coastal-to-inland, the probability of a sharknado occurring conditional on this event can be written as:

$$P(\text{sharknado} \mid \text{tornado is coastal to inland}) = P(\text{shark is present}) \times P(\text{tornado is strong enough to lift a shark} \mid \text{tornado is coastal to inland})$$

The latter case is likely the better answer to our question, as we are worried about the possibility of sharknados occurring when there is a coastal to inland tornado, not when there is a tornado anywhere. As such, the following analysis will focus on this conditional framework.

To obtain a conservative upper bound on this probability, we assume that $P(\text{shark is present}) = 1$. In other words, we assume that whenever a coastal-to-inland tornado occurs, a shark is present near the coast and available to be lifted. While this assumption is unrealistic, it provides an upper bound on the probability of a sharknado. This strengthens the conclusion if the resulting probability remains small.

To begin, we classify each tornado as coastal-to-inland, inland-to-coastal, or neither. While this could be simplified to a binary classification (coastal-to-inland versus not coastal-to-inland), we include the inland-to-coastal category for completeness and to allow comparison between movement directions.

The classification threshold for “coastal-to-inland” tornadoes requires that the tornado’s starting location be within 10,000 meters of the nearest coast and that its ending location be farther from the coast than its starting location. Similarly, tornadoes are classified as “inland-to-coastal” if their starting location is more than 10,000 meters from the nearest coast and their ending location is closer to the coast than their starting location.

how are you determining which direction from the coast? away from the coast could mean going into the ocean rather than on land

Tornado Type	Starting Distance from Coast	Ending Distance from Coast	Criteria
Coastal → Inland	10,000 m	> Starting distance	Moves away from coast
Inland → Coastal	> 10,000 m	< Starting distance	Moves toward coast

Tornado Type	Starting Distance from Coast	Ending Distance from Coast	Criteria
Other	Could be either	Could be either	Doesn't meet the 2 criteria for either group

Coastal to Inland | Inland to Coastal

include transition into the next section
also, overall would like to see some outside research on methods and why selecting this one

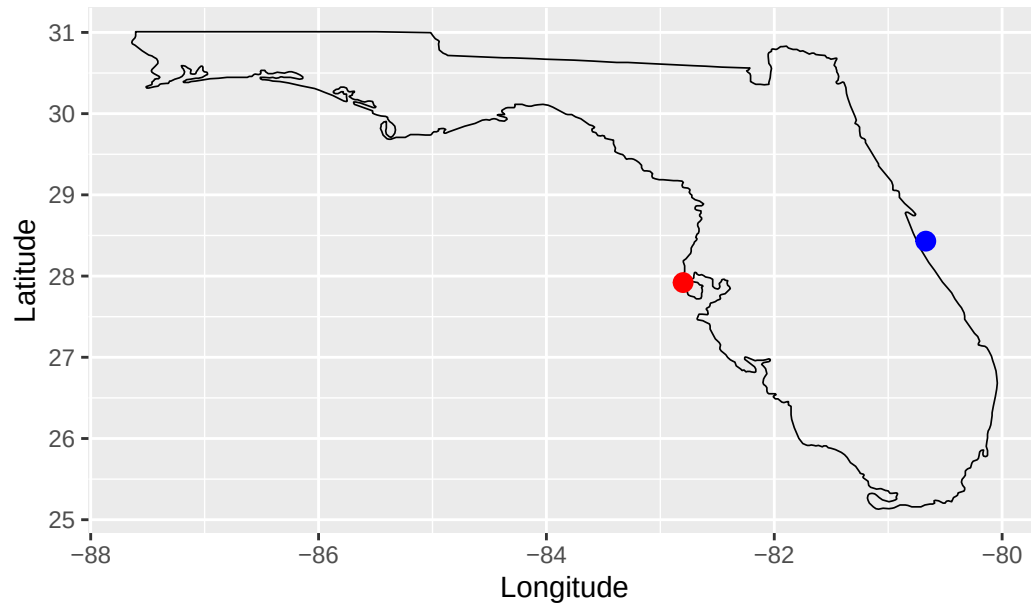
```
florida_tornado <- tornado_paths |>
  filter(mag == 4 & movement == "Coastal_to_Inland")

florida <- map_data("state") |>
  filter(region == "florida")

ggplot() +
  geom_polygon(
    data = florida,
    aes(x = long, y = lat, group = group),
    fill = NA,
    color = "black",
    linewidth = 0.3
  ) +
  geom_point(
    data = florida_tornado,
    aes(x = slon, y = slat),
    color = "red",
    size = 3
  ) +
  geom_point(
    data = florida_tornado,
    aes(x = elon, y = elat),
    color = "blue",
    size = 3
  ) +
  labs(
    title = "Only Possible Sharknado Tornado Found",
    x = "Longitude",
```

```
y = "Latitude"  
)
```

Only Possible Sharknado Tornado Found



5 Results

introduce the section

Now we need to compute the probabilities for each tornado type that we have classified. This can be done quite simply by dividing the number of tornadoes that we have classified to each tornado type by the total number of tornadoes. The results are summarized in the following table:

Tornado Type	Number of Tornadoes	Probability
Coastal to Inland	417	0.93%
Inland to Coastal	16535	36.96%
Other	27786	62.11%

As shown, there are far fewer coastal-to-inland tornadoes compared with inland-to-coastal and other tornadoes. This relatively small amount reflects the strict requirement that the tornado must start within a short distance of the coast to be classified in this category.

Lastly, we estimate the probability that a tornado is strong enough to lift a shark, conditional on it being coastal-to-inland. According to NOAA Fisheries (2025), great white sharks can weigh up to 4,500 lbs. Similarly, Quach (2025) reports that the average car weighs approximately 4,419 lbs, suggesting that great white sharks are of comparable mass to cars. From Otero Property Adjusting & Appraisals (2025), EF3 tornadoes are “likely to toss cars and trucks,” whereas EF4 tornadoes are “very likely to toss cars many feet.” This indicates that at least an EF4 tornado would be required, since a sharknado would require sharks not only to be lifted, but also transported a substantial distance inland rather than merely displaced. To compute this probability, we first have to define the number of tornadoes for each combination of magnitude m and movement type t , call this $N_{m,t}$. The probability of a tornado having magnitude m given movement type t is the number of tornadoes with magnitude m and movement type t divided by the total number of tornadoes with movement type t .

also mention that they have to be pulled out of water - is that possible to calculate with water being an added effect?

$$P(\text{Magnitude} = m | \text{Movement} = t) = \frac{N_{m,t}}{\sum_m N_{m,t}}$$

For our specific probability we are interested in this probability where $m = 4$ or 5 , and $t =$ coastal to inland.

Computing this probability we get:

$$\frac{1 + 0}{417} \approx 0.24\% \quad (1 \text{ from magnitude } 4, 0 \text{ from magnitude } 5)$$

double check how the math notation is interpreting your text visually since underscores turn the following letters or symbols into a subscript. Also what is the ending probability numerically other than converging to zero?

It is important to note that even this estimate represents a conservative upper bound. We assume that a shark is present along the coast whenever a coastal-to-inland tornado occurs, and that an EF4 tornado is sufficient to lift a shark from the ocean onto land. These assumptions are deliberately favorable to the occurrence of a sharknado. Relaxing these large assumptions would cause this probability to converge even closer to 0.

6 Conclusion

The analysis demonstrates that even under highly favorable assumptions, the probability of a sharknado occurring remains extremely low. By breaking down the event into a series of necessary events, each with a very small likelihood, the combined probability becomes effectively zero. Notably, the assumption that sharks are always available along the coastline and that tornadoes of sufficient magnitude can lift them inland represents a significant overestimation, reinforcing that the calculated probability serves as a conservative upper estimate, rather than a realistic expectation.

Additionally, the probabilities seem to be calculated based on tornadoes already having happened

This extremely low probability is to be expected as we are forcing many rare events to occur simultaneously. When this happens, the overall probability decreases dramatically. This idea extends beyond the concept of sharknados and is relevant for other concerns where many rare events need to occur together. In these scenarios, the overall likelihood will drop dramatically as the sharknado probability did.

is it still non-zero? How can the probabilities be compared?

Future work could build upon this analysis by incorporating more realistic environmental constraints, such as the distribution of shark locations and the physics-based modeling of lifting forces required for moving sharks. Ultimately, while sharknados are far from realistic, the analytical approach used in this study shows the power of probabilistic reasoning in exploring even the most unusual problems.

also for discussion is weather trends becoming more extreme

References

- NOAA Fisheries. 2025. “White Shark | NOAA Fisheries. NOAA.” February 19. <https://www.fisheries.noaa.gov/species/white-shark>.
- Otero Property Adjusting & Appraisals. 2025. “Can a Car Be Picked up by a Tornado?” November 6. <https://oteroadjusting.com/can-a-car-be-picked-up-by-a-tornado/>.
- Quach, Jacqueline. 2025. “How Much Does the Average Car Weigh? AutoInsurance.com.” April 29. <https://www.autoinsurance.com/guide/average-car-weight/>.
- “Storm Prediction Center Maps, Graphics, and Data Page.” n.d. Accessed February 23, 2026. <https://www.spc.noaa.gov/wcm/#data>.

A Draft: Data Documentation

A.1 “Storm Prediction Center Maps, Graphics, and Data Page” (n.d.)

```
library(tidyverse)

tornado_full_data <- read.csv("1950-2024_actual_tornadoes.csv")

tornado_paths <- tornado_full_data |>
  distinct(slat, slon, elat, elon, .keep_all = TRUE)
```

The data were collected by NOAA National Weather Prediction Service.

The data was collected simply for the purpose of historical records and public safety, so that there can be research into these tornadoes.

The data is about tornadoes, showing where they occur and how they travel.

The data was collected from 1950-2024, and shows the date each tornado occurred as a column in the dataset.

The data was collected all around the United States. This includes (potentially) all states.

Reports from storm spotters and what the NWS considers reputable sources are submitted to the Storm Data publication by National Weather Service field offices.

The data is a simple CSV file with each row representing one tornado and each column being a different descriptor of the tornado.

The NOAA is a reputable known source for these types of data so there should be no data quality issues.

Some suggested data analysis methods include probability estimation based on magnitude and/or distance travelled inland or towards the coast

https://www.spc.noaa.gov/wcm/data/SPC_severe_database_description.pdf

<https://data.cnra.ca.gov/dataset/shark-incident-database-california>

Tornado Data Variable Descriptions

Variable Name	Description	Type	Possible Values
yr	Year Tornado Occurred	Integer	1950-2024
date	Exact Date Tornado Occurred	Date	01-01-1950
st	State Abbreviation Where Tornado First Touched Down	Categorical	Any US State
mag	EF Scale Magnitude of the Tornado	Integer	0-5 or -9 M
slat	Latitude Where the Tornado First Touched Down	Numeric	US Latitude
slon	Longitude Where the Tornado First Touched Down	Numeric	US Longitude
elat	Latitude Where the Tornado Ended	Numeric	US Latitude
elon	Longitude Where the Tornado Ended	Numeric	US Longitude
len	Tornado Path Length (Miles)	Numeric	0-250 Miles
wid	Tornado Path Width (Yards)	Numeric	0-3500 Yards

i Note

You should reference these documents/web pages formally using Zotero/bibtex. Ideally your documentation would be a bit more structured/formal and have a more varied structure than “The data is/was/were”. Use the same tense throughout a paragraph, please.

i Note

How many of the important values you would want to use are missing? Is there any relationship between missingness? e.g. smaller tornadoes might be less likely to have observed values. Is there a difference in which storms are recorded over time? (e.g. at some point NOAA changed and allowed tornadoes that were radar indicated to be recorded alongside those which were witnessed.)

```
shark_attacks <- read.csv("SharkAttacks.csv") |>
  select(IncidentNum, Date, County, Location, Species, Longitude, Latitude, Comment)

shark_attacks <- shark_attacks |>
  select(IncidentNum, Date, County, Location, Species, Longitude, Latitude, Comment)

unique(shark_attacks$Species)
```

```
[1] "White"      "Hammerhead" "Blue"        "Killer Whale" "Unknown"
[6] "blue"       "Blue*"      "Mako"        "Leopard"      "Sevengill"
[11] "Salmon"     "Thresher"   ""
```



```

shark_attacks_not_sharks <- shark_attacks |>
  filter(!Species %in% c("White", "Hammerhead", "Blue", "blue", "Blue", "Blue*", "Leopard",

shark_attacks_actual_sharks <- shark_attacks |>
  filter(!Species %in% c("Killer Whale", "Unknown"))

unique(shark_attacks_actual_sharks$Species)

```

```

[1] "White"      "Hammerhead" "Blue"       "blue"       "Blue*"
[6] "Mako"       "Leopard"    "Sevengill"  "Salmon"     "Thresher"
[11] ""

```

```

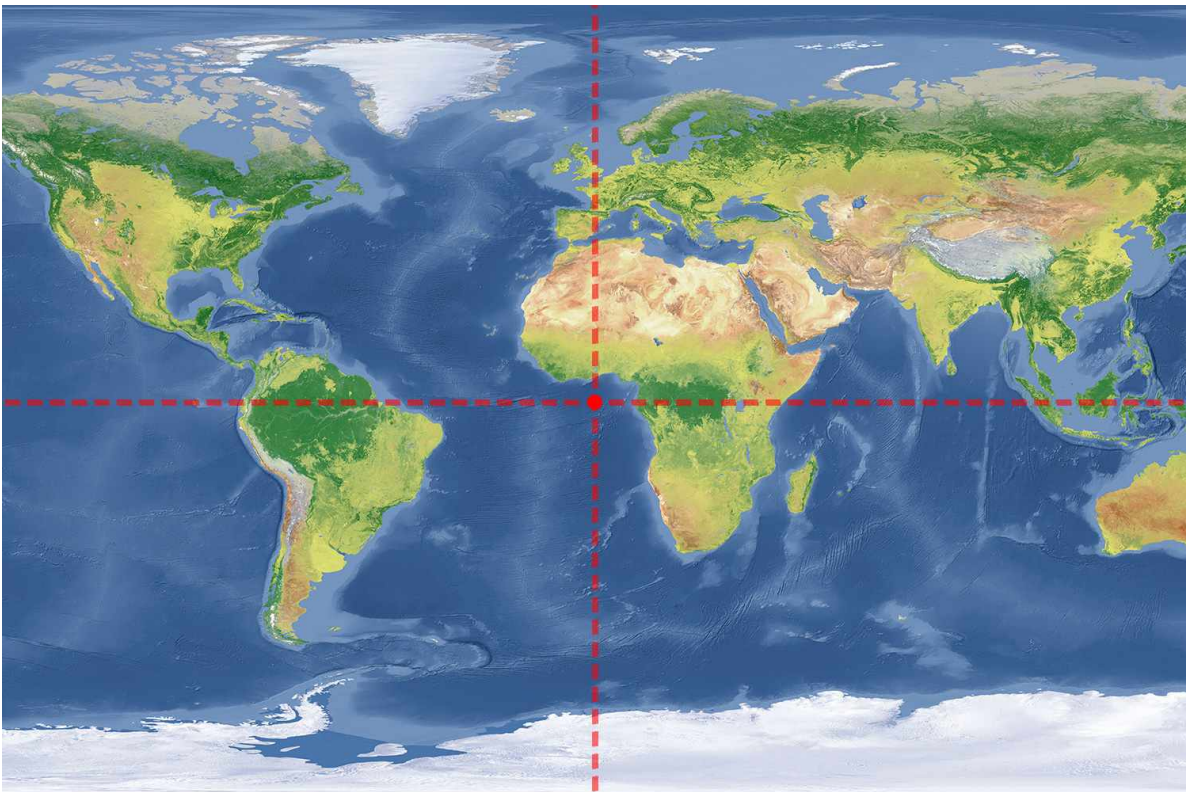
view(shark_attacks_actual_sharks)

```

B Methods

A table summarizing the tornado data from the NOAA and their possible values is below:

In the original US tornado data set from the NOAA, many tornadoes have missing ending latitudes and longitudes, specifically they were marked as “0” instead of “NA”. This can’t be right however since the 0 latitude line and the 0 longitude line do not intersect the US anywhere. The red dotted lines below show 0 latitude and 0 longitude.

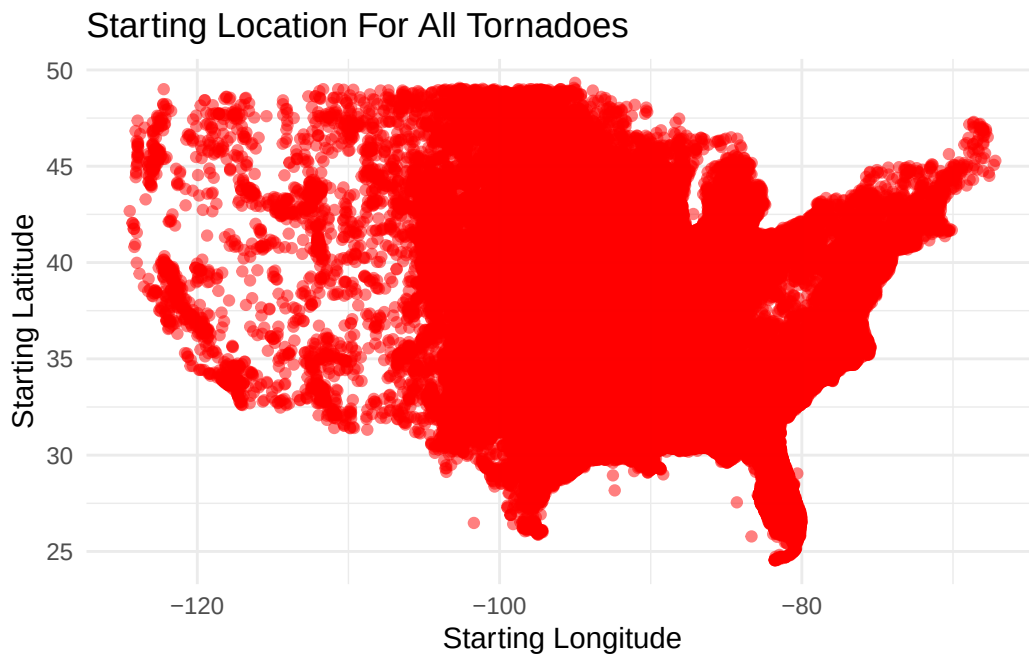


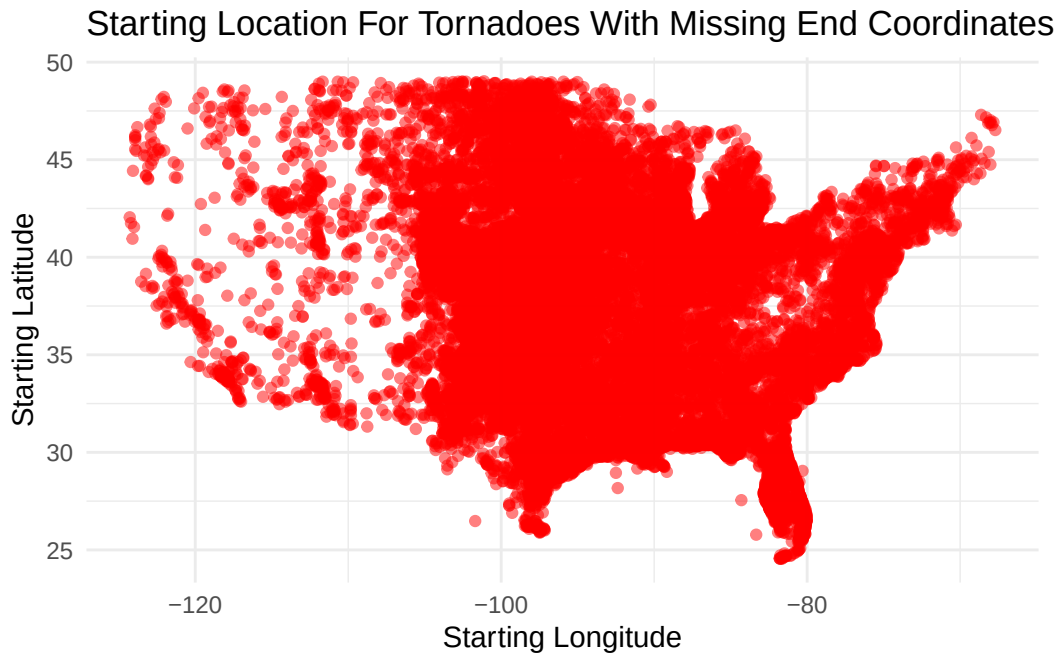
As such, these tornadoes in the data set are taken to be rows with missing values. It should be examined whether these tornadoes have some sort of trend relative to the other tornadoes in the data set, so that it can be determined if they can be removed without worry. Specifically, there are two possible concerns: whether tornadoes in a certain area have a significantly higher chance of missing end values, and whether tornadoes of different magnitudes are more likely to have missing end coordinates than others. To examine the first concern, two figures can be created. One of these shows the location of all tornadoes in the data set, and the other

Tornado Data Variable Descriptions

Variable Name	Description	Type	Possible Values
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elat	Latitude Where the Tornado Ended	Numeric	US Latitude
elon	Longitude Where the Tornado Ended	Numeric	US Longitude
len	Tornado Path Length (Miles)	Numeric	0-250 Miles
wid	Tornado Path Width (Yards)	Numeric	0-3500 Yards

shows the location of only the tornadoes that have a missing value for the ending latitude or longitude.



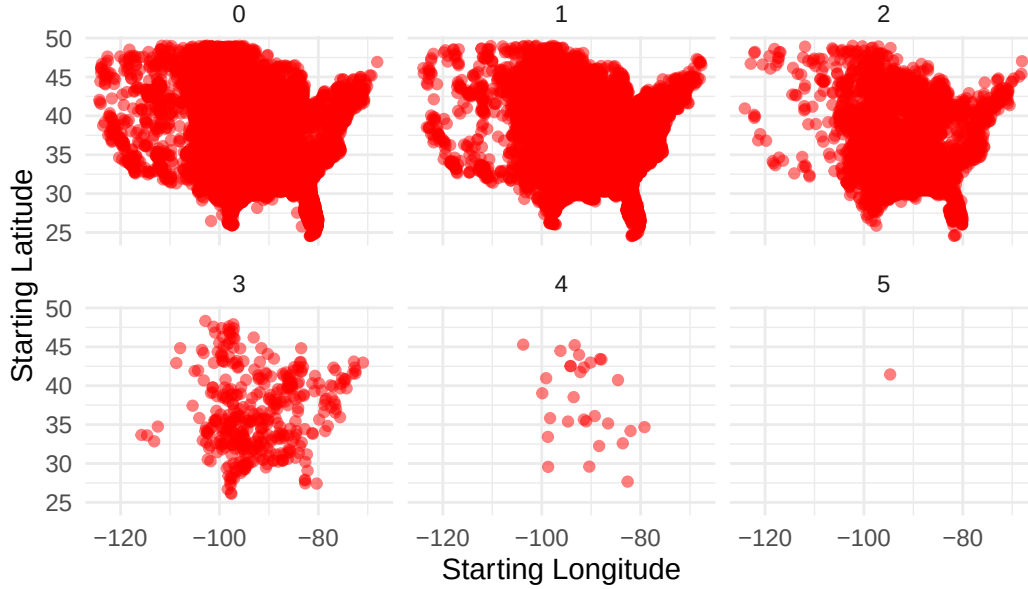


There does not appear to be any trend in the missing end coordinates. The missing end coordinates plot does not appear to have a different trend in location than the plot with all tornadoes. This suggests that our first concern is alleviated.

For the second concern regarding the magnitude, a plot can be created for each magnitude level and examine if there is any difference in the incidence of missing end coordinates depending on the magnitude.

```
library(ggtext)
ggplot(
  tornado_paths |>
    filter(elat == 0 | elon == 0),
  aes(x = slon, y = slat)
) +
  geom_point(color = "red", alpha = 0.5) +
  facet_wrap(~ mag) +
  labs(
    title = "Starting Location For Tornadoes With Missing End Coordinates (by Magnitude)",
    x = "Starting Longitude",
    y = "Starting Latitude"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_markdown()
  )
)
```

Starting Location For Tornadoes With Missing End Coordinates



From these plots, it can be seen that none of the magnitudes show a large difference in location on the map. This alleviates the second concern as well, and as such, we can remove these tornadoes without worry about affecting our inference on the probability of a sharknado.

To examine the potential risk of a sharknado, we compute the probability of such an event occurring. From the perspective of any tornado occurring in the USA, this probability can be expressed as the product of several individual probabilities:

$$P(\text{sharknado}) = P(\text{shark is present}) \times P(\text{tornado is coastal to inland}) \times P(\text{tornado is strong enough to lift a shark})$$

Alternatively, if we focus specifically on tornadoes that are coastal-to-inland, the probability of a sharknado occurring conditional on a tornado being coastal to inland can be expressed as:

$$P(\text{sharknado} \mid \text{tornado is coastal to inland}) = P(\text{shark is present}) \times P(\text{tornado is strong enough to lift a shark} \mid \text{tornado is coastal to inland})$$

The latter case is likely the better answer to our question, as we are worried about the possibility of sharknados occurring when there is a coastal to inland tornado, not when there is a tornado anywhere. As such, the following analysis will focus on the latter case.

To obtain a very conservative upper estimate of this probability, we assume that the probability a shark is present is 100%. In other words, we assume that whenever a coastal to inland tornado occurs, a shark is present near the coast and available to be lifted. While this assumption is not

necessarily realistic, it provides a conservative upper bound on the probability of a sharknado. This will strengthen the evidence when the probability of a sharknado is still quite low.

See [?@fig-sharknado](#) for a flowchart illustrating these necessities.

Change to results here

To start, we must classify each tornado as being coastal to inland, inland to coastal, or neither. This could be done as just coastal to inland or not coastal to inland, but for thoroughness, we included another group for inland to coastal as well to compare.

The threshold used to decide upon “coastal to inland” was if the starting distance of the tornado was within 10000 meters of the closest coast and the ending distance of the tornado was further from the closest coast than the starting distance was. Likewise, tornadoes were classified into the “inland to coastal” group if they were more than 10000 meters from the closest coast and the ending distance of the tornado was closer to the coast than the starting distance.

Tornado Type	Starting Distance from Coast	Ending Distance from Coast	Criteria
Coastal → Inland	10,000 m	> Starting distance	Moves away from coast
Inland → Coastal	> 10,000 m	< Starting distance	Moves toward coast
Other	Could be either	Could be either	Doesn't meet the 2 criteria for either group

Coastal to Inland | Inland to Coastal

Change to results here

Now we need to compute the probabilities for each tornado type that we have classified. This can be done quite simply by dividing the number of tornadoes that we have classified to each tornado type by the total number of tornadoes. The results are summarized in the following table:

Tornado Type	Number of Tornadoes	Probability
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Inland to Coastal	16535	36.96%
Other	27786	62.11%

As shown, there are far fewer coastal-to-inland tornadoes compared with inland-to-coastal and other tornadoes. This relatively small amount reflects the strict requirement that the tornado must start within a short distance of the coast to be classified in this category.

Lastly, we must compute the probability the tornado is strong enough to lift a shark given that the tornado is coastal to inland. Based on NOAA Fisheries (2025), we see that great white sharks can be up to 4500 lbs. Based on Quach (2025), we see that cars weigh on average 4419 lbs. This suggests that the weight of these great whites is comparable to cars. From Otero Property Adjusting & Appraisals (2025), we see that EF3 tornadoes are “likely to toss cars and trucks,” but EF4 tornadoes are “very likely to toss cars many feet.” This suggests that at least a magnitude 4 tornado would be needed, as for a sharknado to occur, we need the sharks to be picked up and tossed quite far onto land, not just to be picked up and tossed slightly. To compute this probability, we first have to define the number of tornadoes for each combination of magnitude m and movement type t , call this $N_{m,t}$. The probability of a tornado having magnitude m given movement type t is the number of tornadoes with magnitude m and movement type t divided by the total number of tornadoes with movement type t .

$$P(\text{Magnitude} = m | \text{Movement} = t) = \frac{N_{m,t}}{\sum_m N_{m,t}}$$

For our specific probability we are interested in this probability where $m = 4$ or 5 , and $t =$ coastal to inland.

Computing this probability we get:

$$\frac{1 + 0}{417} \approx 0.239808153477218\% \quad (1 \text{ from magnitude } 4, 0 \text{ from magnitude } 5)$$

It is important to note that even this very low probability is a conservative upper estimate. We assume that a shark is present along the coast whenever a coastal-to-inland tornado occurs and that a magnitude 4 tornado could lift a shark from the ocean onto land. These assumptions strongly favor the occurrence of a sharknado, since magnitude 4 tornadoes have only been observed lifting cars (objects of similar weight to sharks) several feet off the ground. [enter a source here later probably for some of this information]

i Note

I think you mostly just need to make the writing here a bit more formal and academic, e.g. “Consider that tornadoes with a magnitude lower than EF3 are unlikely to have the wind speed necessary to lift a shark large enough to truly create a sharknado event¹.”

Sources are good too, though.

Your numbers should be dynamically computed – use R’s ability to plug numbers into equations using ‘r ’ but with backticks instead of single quotes. Pull the values directly from the data.

Talk your audience through the results section. Consider highlighting main conclusions in a box or something that’s visually different from the wall-of-text.

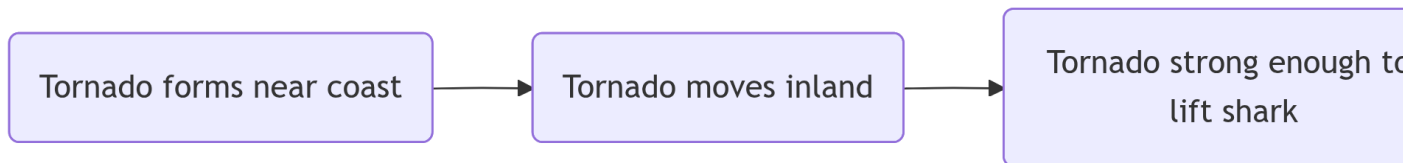
¹One assumes that a shark smaller than 12” would be unlikely to be sufficiently terrifying as to

C Draft: Intro/Conclusions

D Introduction

Extreme weather events have long piqued public and scientific interest, especially when their consequences can be extremely dangerous for human life. Among these, the concept of a “sharknado,” a tornado capable of lifting sharks from the ocean and throwing them inland, represents an exaggerated but intriguing scenario. While such an event is widely seen as fictional, evaluating a sharknado’s plausibility provides an opportunity to apply probabilistic reasoning within real-world constraints.

This project aims to estimate the probability of a sharknado occurring by breaking up the phenomenon into a sequence of events that need to occur. In the methods section, we outline the sequence of these probabilities. These include the probability of a tornado forming near a coastal region and moving inland, the probability that such a tornado reaches a sufficient intensity to pick up and move a shark, and the assumption that a shark is present at the exact location required. To ensure a conservative upper estimate for this probability, several assumptions are intentionally biased in favor of the event occurring, such as a constant availability of sharks along coastlines and the ability for high magnitude earthquakes to pick them up and transport them inland.



i Note

Label, caption, cite in the paragraph.

Using a combination of probabilistic reasoning and simplifying assumptions, the analysis synthesizes these individual components into an overall estimate for the probability of the occurrence of a sharknado. These results provide insight not only into the probability of a sharknado, but also into how compounding rare events together quickly diminishes the overall probability.

E Conclusion

The analysis demonstrates that even under highly favorable assumptions, the probability of a sharknado occurring remains extremely low. By breaking down the event into a series of necessary events, each with a very small likelihood, the combined probability becomes effectively zero. Notably, the assumption that sharks are always available along the coastline and that tornadoes of sufficient magnitude can lift them inland represents a significant overestimation, reinforcing that the calculated probability serves as a conservative upper estimate, rather than a realistic expectation.

This extremely low probability is to be expected as we are forcing many rare events to occur simultaneously. When this happens, the overall probability decreases dramatically. This idea extends beyond the concept of sharknados and is relevant for other concerns where many rare events need to occur together. In these scenarios, the overall likelihood will drop dramatically as the sharknado probability did.

Future work could build upon this analysis by incorporating more realistic environmental constraints, such as the distribution of shark locations and the physics-based modeling of lifting forces required for moving sharks. Ultimately, while sharknados are far from realistic, the analytical approach used in this study shows the power of probabilistic reasoning in exploring even the most unusual problems.

Note

This is reasonable – you may want some sort of citation to the movie and/or an analysis of similar events/rumors...

<https://www.youtube.com/shorts/Rzdzay13RLM>

[https://www.ny-](https://www.ny-times.com/2017/08/28/us/politics/shark-hurricane-harvey-rumors.html)

[times.com/2017/08/28/us/politics/shark-hurricane-harvey-rumors.html](https://www.ny-times.com/2017/08/28/us/politics/shark-hurricane-harvey-rumors.html)

<https://www.youtube.com/watch?v=a5wRZdk9bSI>

https://en.wikipedia.org/wiki/Rain_of_animals (cite the sources themselves rather than the article unless citing the existence of the phenomenon)

<https://www.nj.com/news/2025/08/shark-crashes-karaoke-night-at-jersey-shore-bar-during-hurricane-erin-flooding.html>

<https://www.pastemagazine.com/tech/memes/hurricane-shark-meme-real-climate-change-florida> - Contrary to this article, the meme well pre-dates sharknado - I remember photoshopped pics of sharks in the tunnels under downtown Houston during Tropical Storm Alison circa 2001.

<https://www.youtube.com/watch?v=2g2MLufKBUQ>

F Draft: Executive Summary

The purpose of this report is to examine the probability of a “sharknado” by enforcing the real world constraints of such a phenomenon. The primary objective is to estimate the likelihood of a sharknado by sequencing the probability of all the events that must occur together, and evaluating them using available tornado data and reasonable assumptions.

The methodology starts by addressing missing values in the dataset. Specifically, there are many tornadoes in the original data set that have missing values for the ending latitude or longitude. These tornadoes are excluded from the analysis. This is okay to do as the tornadoes with missing values don’t appear to have any bias in their location. The section continues by describing the probabilities that are sequenced together.

These are:

- 1) The probability that a shark is present for the tornado to pick up
- 2) The probability that a tornado is coastal to inland
- 3) The probability that the tornado is strong enough to lift a shark.

A key assumption made is that sharks are always available to be picked up when a coastal to inland tornado occurs. This assumption creates a great overestimate in the probability as it very much favors the occurrence of a sharknado. This ensures that if the resulting probability is still extremely small, then the sharknado risk is essentially nonexistent.

In the results, the probabilities are computed. It is emphasized that the real world constraints for a sharknado to actually happen cause the probability to be very small. This reflects the idea that sharknados are simply a product of fiction, and have effectively no risk in real life.

Based on the findings of this analysis, no action needs to be taken to adjust insurance products for the risk of sharknados. The probability of such an event happening is effectively zero, and as a result, any expected claims would also be zero. Time and resources are better allocated towards models accounting for other more likely catastrophic events such as tsunamis, hurricanes, or standard tornadoes.